BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. Pencil

Embedded Systems

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before turning to the answers.

1. Name the three CPU characteristics that can affect performance.

2. What does the Memory Address Register (MAR) store? What does the Program Counter (PC) store?

3. What is one difference between ROM and RAM?

4. A 40 x 20 bitmap uses a 4-bit colour depth. Calculate its uncompressed pixel-data size in bytes. Why would lossless compression be needed for an editable source-code file?

Today's objectives

Define an embedded system and recognise examples in everyday devices. Describe key characteristics of embedded systems using OCR J277 language.

Key vocabulary - Read, cover, recall

Embedded system
A computer built into a larger device to perform a specific function.

Dedicated function
A particular task for which a system has been designed.

Sensor
An input device that detects or measures a physical condition.

Actuator
An output device that produces a physical action, such as switching or moving a mechanism.

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. P1 / P2

Learn and practise

New learning - Four key ideas

DEDICATED JOB	A computer built into a larger device for one purpose.
INPUT > PROCESS > OUTPUT	Sensor -> controller decision -> motor or signal.
SOFTWARE / MEMORY	Firmware in non-volatile memory; RAM for working data.
SUITABLE FEATURES	Often low power; some require a timely response.

EMBEDDED CONTROLLER: INPUT -> PROCESS -> OUTPUT

INPUT	PROCESS	OUTPUT
Water sensor	Wash program	Drum motor

The controller is built into a larger washing machine.

I Do - Worked example

A washing machine uses a processor to control its wash cycle. Explain why it is an embedded system, identify a sensor input, and name an output. [4 marks]

1. The washing-machine controller has a dedicated job.
2. It is a computer built into a larger device.
3. Input: water-level or temperature sensor.
4. Output: switch the drum motor on or off.

We Do - Guided practice

Thermostat: why embedded, and what input and output?

BLACK PEN - complete your explanation:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes. Page 3 of 4

Think and write

Code First - Predict > Write > Test > Improve

Read the OCR Exam Reference Language code before testing it.

```
temperature = 29
if temperature > 28 then
print("Ventilation on")
else
print("Ventilation off")
endif
```

1. Predict the result at temperature = 29.

2. Change temperature to 22 and predict again.

3. Use limit = 28 and test at temperature = 28.

You Do - Independent exam questions

Q1. State TWO characteristics of an embedded system. [2 marks]

Q2. A car-park barrier uses a vehicle sensor and a motor to lift the barrier when authorised. Explain why its controller is an embedded system, and identify an input and an output. [4 marks]

Q3. A battery-powered smoke alarm uses a sensor to detect smoke and activates a loud sounder when smoke is detected. Explain THREE embedded-system characteristics that make it suitable for this job. [6 marks]

Exit Ticket - True or False?

Answer before checking Page 4.

1. Every embedded system must have a full keyboard and monitor. ☐ T ☐ F

2. A thermostat may contain an embedded controller. ☐ T ☐ F

3. A sensor provides an input, not a physical output. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Mark fairly. Tick correct points and fix any mistakes.

Use the mark allocations; then rewrite one answer in full.

Do Now - Answers

Q1 Clock speed, cache size and number of CPU cores. ☐

Q2 MAR: the address of a memory location being accessed. PC: the address of the next instruction to fetch. ☐

Q3 ROM is non-volatile and typically stores firmware; RAM is volatile and stores currently used programs and data. ☐

Q4 40 x 20 = 800 pixels; x 4 = 3,200 bits; / 8 = 400 bytes. Lossless compression allows the original source code to be restored exactly. ☐

Code First - Answers

1. At 29: "Ventilation on". ☐

2. At 22: "Ventilation off". ☐

3. At 28, condition 28 > 28 is false: Ventilation off. ☐

We Do - Check the model

Built into heating controller [1], dedicated purpose [1], temperature-sensor input [1], heating switch output [1].

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

Explain one false statement using the key ideas.

You Do - Self-mark

Q1 Two features: built into a larger device [1] and a specific job [1]. ☐

Q2 Barrier system has a dedicated controller [2]; vehicle sensor is input [1]; motor is output [1]. ☐

Q3 Smoke alarm: dedicated smoke detection [2]; low power for battery [2]; real-time response for safety [2]. ☐

Try the question again

1. Copy the FULL question you got wrong (or found hardest):

2. In GREEN PEN, write a NEW answer. Include working if needed: